

Optimal Connectivity of Multilayer Networks: Enhancing Robustness and Security

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Abstract—Modern communication systems typically rely on complex topologies in which different devices interact through interconnected networks. While the versatility of multilayering enhances network capability and performance, multiple interlayer connections pose an additional security risk, as they can serve as pathways for malware to spread. Reducing the number of interconnections between layers helps mitigate propagation. However, limiting these connections increases the risk of network isolation due to cyberattacks that compromise communication links. This paper studies the problem, adopting the SIR epidemic model within a two-layer framework. It demonstrates the existence of an optimal number of edges that maximizes the probability of remaining connected while preventing the malware from spreading from one layer to another. We also present an approach for determining the optimal number of edges, based on the weighted scoring method, that considers personal preferences. Extensive simulations using the SIR epidemic model corroborate our findings and assess the impacts of the model parameters on the spread of malware.

Index Terms—Robustness, malware, multilayer network, connectivity.

I. INTRODUCTION

THE widespread adoption and large-scale deployment of ubiquitous computer systems, such as the Internet of Things (IoT), encompassing everything from surveillance cameras to smartphones and industrial cyber-physical systems, should be accompanied by the development of technological solutions capable of mitigating modern security threats [1]. To address the contemporary privacy and security risks posed by malware and skilled cybercriminals, multiple approaches have been employed, ranging from the use of antimalware software and firewalls to the adoption of security policies by users in agencies and companies.

Developing models that accurately represent the propagation of malware in networks is crucial for evaluating responses and enhancing security methods. At the same time, complex and

dedicated networks involving multilayer topologies are becoming increasingly popular in diverse applications. Cyber-physical systems, for example, adopt a two-layer topology, where the lower layer corresponds to physical devices, such as actuators and sensors, while the upper layer involves real users, software agents, and other entities [2].

The problem of propagation across multilayer networks has been investigated in various papers that characterize the dynamic behavior from both macroscopic and microscopic perspectives [3]. In addition, the influence of directionality in complex networks has been addressed in [4], where the authors studied a susceptible-infected-susceptible (SIS) propagation model on two-layer directed multiplex networks. Their work revealed that directionality has a positive impact on the ability to suppress the spread of epidemics. Unfortunately, the applicability to computer networks is minimal as information ordinarily travels in a non-directed way.

From a topological perspective, the authors in [5] assessed the robustness of various complex communication networks against cyberattacks. They proposed a stochastic modeling framework for malware propagation and performed an analysis of random attacks to determine the robustness of different topologies. Additionally, in a novel approach, the work of [6] employed a hypergraph-based model to represent malware propagation over a large-scale wireless network. Hypergraphs can preserve the complete topological information of complex networks, and this study contributes to understanding how to mitigate malware propagation by adjusting the topology.

Moreover, the authors in [7] introduced a model based on Monte Carlo simulations in which two susceptible-infected-recovered (SIR) epidemics propagated simultaneously on a two-layer network and interacted dynamically. They analyzed the function of various factors, including topological parameters, and demonstrated that an increase in the average degree results in a significant growth in the scale and speed of outbreaks. In an alternative work, the researchers in [8] studied the SIR propagation model on an interconnected system with two layers and identified the dynamic behavior of epidemics for both weakly and strongly coupled systems. They found that, in strongly coupled systems, all layers are simultaneously either free of infection or part of an epidemic, while in weakly coupled systems, the disease is epidemic on only one layer despite the interconnections.

In another closely related work, the authors of [9] analyzed the SIS spreading model occurring on two interconnected networks

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and recognized a highly nontrivial behavior of the epidemic process, depending on the strength and nature of the coupling. They demonstrated that a global endemic state might arise in the coupled system, even though the epidemics are unable to propagate on each network separately. Further, they found that this effect is more or less important depending on the strength of the coupling, i.e., the number of links between the two networks.

Evaluating the robustness and vulnerability of multilayer networks is a crucial area of network science, particularly because failures in one layer can trigger cascading failures across all interconnected layers [10]. The problem has been discussed in [11], which presents a method for assessing robustness and vulnerability using three probabilistic parameters: number of nodes, the denseness of network connections, and node failure probability. The paper examines the network behavior under node failures, but does not investigate why a node failed. The cause could be a cyberattack or an environmental disaster – the reason is irrelevant to the analysis.

The challenge of determining the optimal topology for robustness in complex networks has been explored in earlier studies. In [12], the authors analysed the security of a two-layer network against cyberattacks that compromise communication links and calculated the optimal design, along with a lower bound on the number of links, for a specified budget. Their focus was on the effects of network link failures, without addressing the mechanics of the attack or malware behavior on the network. A similar investigation is presented in [13], which assesses the robustness of a cyber-physical system composed of two interconnected networks under random attacks on network nodes. The research considers that a robust network can withstand a higher percentage of node removal before failing, characterizes robustness under a uniform distribution of interedges, and identifies the optimal interedge allocation. While this study explores the effects of node removal caused by attacks or neutralization, it does not delve into the mechanisms or operational details of the attacks themselves or how nodes are specifically targeted.

An interesting approach is presented in [5], which studies the robustness and vulnerability of various complex network topologies under malware attacks. The paper uses the SIS (susceptible-infected-susceptible) propagating model to show that, topologically, the random network is the most robust, while the random geometric network is the most vulnerable. Specifically, the work measures vulnerability as the expected number of infected nodes, and robustness as the inverse of vulnerability. Although the paper investigates complex topologies, it does not explicitly address multilayer networks, nor examine the loss of connectivity due to node or link failures.

Another insightful study investigates the spread of malware in [14] using the SIS (susceptible-infected-susceptible) model within a two-layer framework. The paper analyzes the effects of structural and dynamic disturbances on malware propagation. While it effectively models malware behavior within the complex network, it cannot evaluate robustness or vulnerability

under a malware attack because the network perturbations are introduced from an external source.

In this paper, we examine malware propagation in a double-layer framework and the structural disruption in the network directly caused by malware spread itself. Specifically, we model malware behavior and explore the trade-off between robustness and vulnerability. Here, a network is considered robust if it maintains connectivity between the two layers after an epidemic process reaches a stationary state. At the same time, vulnerability is defined as the likelihood of cross-layer contamination. Therefore, adding interlayer edges enhances robustness, making the network harder to disconnect, but it also increases vulnerability, or equivalently, weakens security by creating new pathways for contagion.

Formally, the optimal trade-off between robustness and security is achieved at a critical number of interlayer edges – namely, the point that maximizes the joint probability of maintaining interlayer connectivity while preventing malware from spreading between layers.

This article contributes to the literature by developing a SIR epidemic model that takes place on a two-layer interconnected network, representing malware propagation at two levels: an external level and a local network level. We note that the infection originates from the external level, which is exposed to attacks, and can be defined in various ways, such as a country network, an ISP domain, and so on. We assume that the malware spreads from the external level to the local network through the edges connecting both levels.

Through Monte Carlo simulations, we investigated the impacts of the number of edges on both the risk of losing connectivity and the probability of infecting the local network. Furthermore, we demonstrated that there is an optimal number of edges that maximizes the probability of remaining connected and not infecting the local network. Therefore, this number represents the optimal trade-off between robustness and vulnerability and depends on various factors, including the parameters of the propagation model and the initial percentage of infected on the external level. Finally, we can frame the contributions of this article as follows:

- We propose a two-level framework, representing an external and local network, for investigating the increase in the number of edges that, on the one hand, improves robustness while, on the other hand, broadens the pathway for invasion;
- Through Monte Carlo simulations, we characterized the robustness and vulnerability of the framework as a function of the number of connecting edges and demonstrated the existence of an optimal number of edges that maximizes the probability of remaining connected while not infecting the local network;
- We present an approach for selecting the optimal number of edges, based on the weighted scoring method, that considers personal preferences towards robustness or vulnerability.

The organization of this paper is as follows. In the next section, we formally define probabilistic measures of network security

and robustness and demonstrate that there exists a number of interlayer edges that balances them; we further frame the behavior of the two-layer network as a phase-transition model at a critical threshold. In Section III, we present the proposed framework and discuss the methodology and simulation design. Experiments are conducted, and the results are thoroughly discussed in Section IV. Further, in Section V, we introduce a methodology for selecting the optimal number of edges using the weighted score method, which accounts for personal preferences. Finally, Section VI offers closing remarks.

II. PRELIMINARY CONCEPTS AND DEFINITIONS

In this section, we define security and robustness as two probabilistic functions. In addition, we demonstrate that a critical number of edges, N^* , exists and optimizes the trade-off between robustness and security. Finally, we describe the two-layer network behavior as a phase transition model between connectivity-limited and security-limited regimes.

A. Mathematical Formulation of Robustness and Security Network Measures

Let $G = (G_1, G_2, E_{INT})$ denote a two-layer network, where $G_1 = (V_1, E_1)$ is the external network, $G_2 = (V_2, E_2)$ is the local network, and $N = |E_{INT}|$ is the number of interlayer edges. In this framework, we assume that an infection originates on the external network (G_1) and spreads through the interlayer edges to the local network (G_2). Finally, we define two independent, discrete probability functions determined by N that govern the security and robustness of the network:

- 1) *Probabilistic measure of network security* P_I : Let $P_I(N)$ be the probability that the local network remains free of infection after an epidemic process reaches a stationary state. Because adding interlayer edges introduces alternative paths for malware to cross layers, $P_I(N)$ is a strictly decreasing function of connectivity N .
- 2) *Probabilistic measure of robustness* P_L : Let $P_L(N)$ be the probability that the network maintains connectivity between the two layers after the epidemic attains a steady state. In other words, $P_L(N)$ denotes the probability that at least one interlayer edge remains free of infection after the epidemic reaches a stationary state. Because increasing the number of interlayer edges reduces the risk of connectivity loss, the probability $P_L(N)$ strictly increases with N .

B. The Joint Probability Formulation for Optimization

Designing a robust and secure two-layer network involves reconciling two fundamentally opposing topological requirements. On the one hand, enhancing structural integrity requires growing the number of interlayer edges. On the other hand, containing epidemiological outbreaks implies reducing the pathways through which malware can traverse layers.

To achieve an optimal infrastructure configuration, the network must satisfy both conditions simultaneously. Therefore, we define the joint probability P_R that the network maintains interlayer connectivity while preventing the malware from spreading

from the external to the local layer:

$$P_R = P(I \cap L) \quad (1)$$

Further, according to the conjunction principle of probability theory, we may write:

$$P_R(N) \leq P_I(N) \quad (2)$$

$$P_R(N) \leq P_L(N) \quad (3)$$

Also, N is a positive integer constrained by $1 \leq N \leq N_{MAX}$, where N_{MAX} is the maximum number of distinct interlayer edges. Let $|E_1|$ and $|E_2|$ denote the number of edges in networks G_1 and G_2 , respectively. Assuming there are no multiple edges between the same pair of nodes, we have:

$$N_{MAX} = |E_1| \cdot |E_2| \quad (4)$$

Therefore, the domain of $P_R(N)$ is a finite set of integers:

$$\{1, 2, 3, \dots, |E_1| \cdot |E_2|\}$$

Because $P_R(N)$ is evaluated exclusively at these values, the possible output of $P_R(N)$ is a finite set of real numbers:

$$S = \{P_R(1), P_R(2), P_R(3), \dots, P_R(|E_1| \cdot |E_2|)\}$$

Since S is a finite, non-empty set of real numbers, it contains a maximum element, which is greater than or equal to all other values in S . This maximum value is the optimal trade-off between robustness and security and is reached at a critical number of interlayer edges N^* .

C. Dynamic Trade-Off Between Robustness and Security

To better understand how the two-layer network behaves, we can gradually add interlayer edges to track the dynamic evolution of robustness and security.

Starting with only one interlayer edge, the network is highly fragile; if it loses its single pathway, the two layers are completely isolated. Adding the first few interlayer edges significantly improves robustness, providing a substantial structural benefit. However, once the layers are well connected, the contribution of additional edges to robustness diminishes, yielding progressively smaller marginal gains.

Conversely, when there is only one interlayer edge, security is maximum, as the risk of malware crossing layers is minimal. Adding new edges significantly reduces security by providing new infecting routes. As connectivity increases, however, each additional edge yields a smaller reduction in network security.

Let k be the last integer such that $P_L(k) \leq P_I(k)$; then $P_L(k+1) \geq P_I(k+1)$. We can say that k represents a transition threshold beyond which the system changes from a connectivity-limited regime to a security-limited regime.

Thus, we can also frame the behavior of the two-layer network as a phase transition model at the critical threshold k and describe it as follows:

- *The connectivity-limited regime* ($P_I(N) > P_L(N)$): In the sparse-connectivity domain, the epidemic is contained, but the probability of maintaining connectivity is low. Consequently, $P_R(N)$ is strictly bounded by $P_L(N)$, and the system remains fundamentally bottlenecked by its lack of robustness.

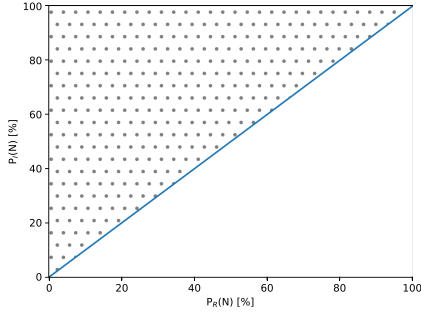


Fig. 1. Graph for $P_I(N) = P_R(N)$. The gray-hatched area indicates the points that satisfy the inequality $P_R \leq P_I$.

- **Phase transition:** As N increases, the system approaches a critical point at which the dominant constraint shifts. The transition manifests as a discrete crossover at an index k .
- **The security-limited regime ($P_L(N) > P_I(N)$):** Beyond the critical threshold, the network is highly connected, and robustness ceases to be the limiting factor. Instead, the dense interlayer connectivity increases the risk of malware propagation. In this regime, the bottleneck is entirely security-driven; the probability $P_R(N)$ is no longer limited by $P_L(N)$ but bounded by the decaying trajectory of $P_I(N)$.

In Fig. 1, we further illustrate the relationship between $P_I(N)$ and $P_R(N)$, with $P_I(N)$ on the vertical axis and $P_R(N)$ on the horizontal axis. The condition given by the (2) means that all valid points $(P_R(N), P_I(N))$ must lie within the gray-hatched area. In addition, since $P_I(N)$ is strictly decreasing, the ceiling bounding $P_R(N)$ decreases as N increases, forcing $P_R(N)$ to attain its maximum at some point in the sequence.

Moreover, it should be noted that the existence of an optimal number of interlayer edges that maximizes $P_R(N)$ is independent of the specific topology of the network and holds for any pair of graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$.

In this work, we study P_L and P_I and identify the optimal balance between them under different conditions through numerical simulations. Achieving reproducible and computationally feasible results requires adopting a tractable topological boundary. Specifically, the simulations are restricted to a two-layer network in which both layers have identical node sets and intralayer connectivity, but differ in the number of interlayer edges.

Additionally, we apply the SIR compartmental model to capture the aforementioned trade-off between robustness and security and validate it as a reliable framework for exploring these competing properties in multilayer networks.

III. PROBLEM DESCRIPTION AND SIMULATION DESIGN

In this section, we describe the model and methodology used to study the problem. Firstly, we provide an overview of cellular automata, the theoretical framework chosen for representing the system. We then provide a detailed explanation of the two-layer network model based on stochastic cellular automata, describing the steps involved in setting up Python software to perform numerical calculations.

A. Overview of Cellular Automata

Cellular automata (CA) were presented by John von Neumann in the 1950s as a class of dynamical systems defined in discrete space and time, comprising an array of cells in n -dimensions, where each cell can be in one of k different states at each time t . The CA framework can describe the spatiotemporal behavior of complex systems and, therefore, can represent the propagation in multilayer networks. Additionally, CA is considered appropriate and has already been adopted for describing malware propagation in computer networks [15].

Formally, the CA can be defined as a 4-uplet:

$$CA = (N, Q, V, f) \quad (5)$$

where, N is the cellular space, Q is the finite set of possible states of the cells, and each cell can be in one state at any discrete time t . Further, each cell may change its state according to a set of transition rules. This set of rules defines the behavior of the cell and depends on its state and the states of its neighbors. Additionally, the neighborhood of the cell must be clearly determined [16].

The state of the cell i at time t is given by $s_i(t)$, and the state of its neighborhood by $s_{vi}(t)$, where V_i stands for the neighborhood of the cell i . Finally, the set of transition rules that define the next state of the cell based on its own state and the state of its neighborhood at any time t is given by f . Therefore, we can write:

$$s_i(t+1) = f(s_i(t), s_{vi}(t)) \quad (6)$$

B. Cellular Automata on Graphs

While the CA framework applies a set of transition rules to all ordered cells with bi-directional links, this work adopts a generalized version of CA, known as *graph cellular automata* (GCA) [17], which relaxes the assumption of a regular structure and allows cells to have arbitrary topologies and different neighborhood configurations.

Formally, a GCA is represented by a 4-uplet:

$$GCA = (\zeta, \sigma, \eta, \tau) \quad (7)$$

where $\zeta = (V, E)$ is a graph with node set V ($|V| = n$), edge set $E \in V \times V$, $\eta : V \rightarrow 2^V$ represents the neighborhood function $\eta(i) = \{j | (j, i) \in E\}$, σ is the set of possible states, and τ denotes the transition rules, and can be expressed as

$$\tau(s_i) : \{s_i\} \cup \{s_j, e_{ji} | j \in \eta(i)\} \mapsto s'_i \quad (8)$$

C. Model Description

We propose a GCA model of malware propagation at two levels: an external level and a local network level. The representation using GCA provides the ability to analyze the phenomena from a spatiotemporal perspective, which is convenient for investigating the influence of topological parameters, including the number of connecting edges, on the system dynamics.

Here, we consider that both the local and external networks can be described by a GCA, each with 32×32 nodes representing the hosts in the network, which can be in one of three different states: susceptible, infected, and removed. In addition, each node connects to eight neighbors, namely the Moore neighborhood, as shown in Fig. 2.

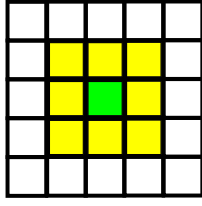


Fig. 2. Cell (green square) and its neighborhood (yellow squares).

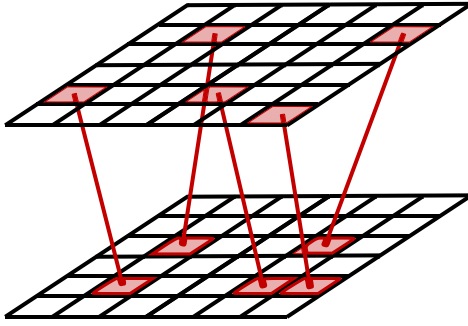


Fig. 3. Topology of a two-layer system GCA architecture, with an external network (upper level) and a local network (lower level). The red squares represent the cells from each corresponding layer that are connected by the internetwork edges (red paths).

As shown in Fig. 3, we study malware propagation at two levels, represented by two GCA connected by internetwork edges.

In this framework, the external network and the local network, represented as two bidimensional GCA, are spatially connected by internetwork connections. The interconnections among different networks occur only between a cell from the external network and another cell from the local network, as illustrated in Fig. 3. We note that there are no overlapping regions, and the interconnected cells, in addition to their neighbors, have an outer neighbor from the adjoining network.

Mathematically, we may describe the framework as follows:

Let the set of network layer indices be $L = \{1, 2\}$.

For each network layer $l \in L$, define the grid cells.

$$V^l = \{(i, j, l) | i, j \in \{1, 2, \dots, 32\}\} \quad (9)$$

The total cell set is $V = V^1 \cup V^2$ and $|V| = 2 \times 32^2 = 2048$.

Within a network l , each cell is locally connected to its eight surrounding cells (Moore neighborhood), so the neighborhood η^l can be written as below:

$$\eta^l(i, j) = \{(i + \Delta i, j + \Delta j, l) | \Delta i, \Delta j \in \{-1, 0, 1\}, (\Delta i, \Delta j) \neq (0, 0)\} \quad (10)$$

The cells on the edges have fewer neighbors – only those that exist inside the grid.

$$\eta^l(i, j) = \{(p, q) \in V^l | |p - i| \leq 1, |q - j| \leq 1, (p, q) \neq (i, j)\} \quad (11)$$

Algorithm 1: Transition Rules of the Cellular Automata.

```

START
INT neighbors_i
REAL beta, gamma
/// Possible states: S, I, R
/// Beta and gamma are the probabilities of getting
    infected and removed, respectively
FOR e  $\in$  elements of the automata DO
    IF state(e) = S THEN
        neighbors_i = number of I neighbors
        FOR X = 1 TO neighbors_i
            r = rand()
            IF r < beta, state(e) = I
        END FOR
    ELSEIF state(e) = I THEN
        r = rand()
        IF r < gamma, state(e) = R
    END IF
END FOR
STOP

```

We define the internetwork edge set as $E_{INT} \subseteq V^{(1)} \times V^{(2)}$. Each element $(u, w) \in E_{INT}$ connects node $u \in V^{(1)}$ with node $w \in V^{(2)}$.

Also, we adopt $|E_{INT}| = i$, i.e., there are exactly i internetwork edges, so that only some nodes receive an extra neighbor. Further, E_{INT} is simple (no duplicate edges) and undirected.

We implemented the malware propagation using the classic SIR epidemic model initially proposed by Kermack and McKendrick [18]. In this model, the population comprises three subgroups: those who are susceptible (S) to infection, those who are already infected (I) and can transmit, and the individuals who are removed (R) and can no longer be infected.

Let us denote $\sigma = \{S, I, R\}$ as the finite set whose elements represent all possible states of the cell and $s_n(t) \in \sigma$ the state of the cell n at time t , we define:

$$s_n(t) = \begin{cases} S, & \text{cell } n \text{ is susceptible at time } t \\ I, & \text{cell } n \text{ is infected at time } t \\ R, & \text{cell } n \text{ is removed at time } t \end{cases}$$

In this model, at each time step, a susceptible cell with an infected neighbor becomes infected with a probability β . The parameter β controls the rate of spread for the malware, affecting the overall aggressiveness of the infection process. Furthermore, infected cells are removed with a probability γ . Once removed, these cells do not recover, are excluded from communication, and no longer transmit data between neighboring nodes. Thus, removing cells that connect to internetwork edges causes the corresponding links to fail permanently.

The transition rules, which determine the dynamic behavior of the SIR model, are defined by the algorithm whose pseudo-code is:

The pseudo-code establishes the dynamic behavior of the CA and corresponds to the classic SIR epidemic model, whose possible state transitions are shown in Fig. 4. In the code, *beta* and *gamma* determine the probabilities of getting infected



Fig. 4. State transition diagram of the SIR epidemic model.

and being removed, respectively. Also, the updating stochastic transition rules are implemented using a uniformly distributed random number generator $rand()$.

Here, the external network can be defined in various ways, such as a country network, a Wide Area Network (WAN), an Internet Service Provider (ISP) domain, and so on. At the same time, the local network can represent a Personal Area Network (PAN), an Enterprise Network, or other types of networks. Besides, the internetwork edges are the legitimate communication links that malware exploits to penetrate from the external domain into the local system.

In this paper, we aim to characterize the robustness and vulnerability of a two-layer network as a function of the number of connecting edges. Hence, we study the following questions.

Given an infection that originates from the external network and propagates to the local network through the interconnecting edges, what are the impacts of the number of edges on both the risk of losing connectivity and the probability of infecting the local network. In other words, we investigate the problem of increasing the number of edges, which, on the one hand, improves robustness while, on the other hand, broadens the pathway for invasion.

D. Computational Implementation of the Model

To accurately examine the problem and answer our questions, we developed a computational model based on GCA to evaluate the dynamic behavior of malware spreading on a two-layer network. The simulation package was written in pure Python programming language and ran on PyPy v7.3.17, a Just-in-Time Python compiler [19], to speed up the code.

In our computational framework, we adopted an object-oriented design to implement the GCA model, where each element or node is represented as an object with its attributes, which correspond to its state and neighborhood. Further, a Python function implements the transition rules described by the Algorithm 1.

It should be noted that the external and local networks can have different β and γ parameters. However, for simplicity of calculation and without loss of generality, we adopt similar β and γ for both layers.

The evaluation of the computational model can be summarized as follows.

- 1) Implement the two-layer network with 1024 nodes each, disposed over a 32x32 grid. The nodes in each layer have the same β and γ parameters;
- 2) For a given number of edges N , connect two randomly chosen nodes from each network. Each edge should be connected to two different nodes, and a node cannot have more than one internetwork edge;

- 3) Assuming an initial fraction of infected nodes I_0 in the external network, the malware spreads from node to node through the internetwork edges to the local network. The dynamic behavior of the model is determined by the transition rules and evaluated in discrete time steps.

Using 15,000 Monte Carlo simulation runs, the program calculates the probability that the local network remains free of infection, namely $P_I(N)$, and the probability that the local network is not attacked and at least one edge remains intact, namely $P_R(N)$. Here, an edge is considered intact when both nodes connected by it remain susceptible by the end of the run. Further, the Monte Carlo analysis is performed for a number of edges N , ranging from 1 to 7.

Since $P_L(N)$ and $P_I(N)$ are defined in the stationary regime of the epidemic process, each simulation is performed over 300 time steps. Preliminary simulations demonstrated that this number of time steps is sufficient for the system to converge to a stationary state, with the average number of S , I , and R stabilizing within an absolute tolerance of 0.01.

Specifically, by calculating $P_I(N)$, we evaluate the impact of N on the vulnerability, i.e., the risk of infecting the local network. In addition, by estimating $P_R(N)$, we determine the impact of N on both robustness and vulnerability, that is, the probability of maintaining connectivity while not infecting the local network. Therefore, assessing the optimal trade-off between robustness and vulnerability consists of finding the value of N that maximizes $P_R(N)$.

In the analysis, we tackle the problem of selection bias by randomly assigning the internetwork edges at each simulation run. This procedure is important because changing the relative positions of the edges could result in different propagation.

Performing multiple Monte Carlo simulations requires a fair amount of processing and computation time. Therefore, we adopted a parallelized code in Python, developed for the simultaneous analysis of different numbers of edges, to significantly reduce execution time.

IV. SIMULATION RESULTS AND DISCUSSION

In this section, we investigate the behavior of $P_I(N)$ and $P_L(N)$ across different values of the parameters beta and gamma, as well as the initial fraction of the infected, I_0 . By running Monte Carlo simulations, we examine the cases and thoroughly discuss the results. Finally, we simulate a hypothetical cyberattack on electric grid infrastructure to demonstrate how our model guides the design of robust, secure, and complex network architectures.

It should be noted that one of the primary objectives of the following simulations is to qualitatively illustrate how the SIR compartmental epidemic model captures the competing concepts of security and robustness in multilayer networks. Moreover, obtaining computationally tractable results requires restricting both the network topology and the model parameters. Accordingly, we consider a two-layer network in which both layers have the same topology, the same number of nodes, and identical parameter values beta and gamma.

TABLE I
 $P_R(N)$ FROM THEORETICAL CALCULATION AND SIMULATION, WITH RESPECTIVE MARGIN OF ERROR Δ

	N=1	N=2	N=3	N=4	N=5	N=6	N=7
$P_R(N)$ (Theoretical)	9.96%	18.94%	27.03%	34.32%	40.89%	46.80%	52.13%
$P_R(N)$ (Simulation)	10.07%	19.09%	27.75%	34.37%	40.39%	47.23%	52.69%
Δ (Simulation)	0.48 %	0.63%	0.72%	0.76%	0.79%	0.80%	0.80%

A. For $\beta=0$, $\gamma=0.1$ and an Initial Fraction of Infected $I_0=0.9$

We analyze this specific case where there is no spreading ($\beta = 0$), which falls outside the model's assumptions. Although this scenario does not meet the criteria of $P_I(N)$ being strictly decreasing, it is included because it allows an analytical solution, offering a useful benchmark for validating our simulation.

For $\beta = 0$, there is no risk of infecting the local network; that is, $P_I(N) = 1$, and consequently, $P_R(N)$ is given by the probability of not all edges being connected to infected nodes, and can be calculated by the following equation:

$$P_R(N) = 1 - \frac{\binom{I}{N}}{\binom{n}{N}} \quad (12)$$

where N is the number of edges, n is the total number of nodes $n = 32 \cdot 32 = 1024$, and I is the number of infected nodes in the external network; $I = I_0 \cdot n \approx 922$. In fact, solving $P_R(N)$ is a combinatorial problem such that for a given number of infected $I = I_0 \cdot n$, find the probability that not all edges are connected to I .

Running the Monte Carlo simulation to estimate the probability P_R essentially represents conducting a series of independent Bernoulli trials. The estimated probability P_R is calculated by:

$$\hat{P} = \frac{\theta}{R} \quad (13)$$

In this context, θ indicates the count of instances wherein at least one interlayer edge remained uninfected, and R denotes the total number of Monte Carlo simulations performed, with $R = 15000$.

The standard error δ measures the extent to which the estimated probability is expected to vary from the true value, accounting for the randomness inherent in the simulation. It can be computed as follows:

$$\delta = \sqrt{\frac{\hat{p}(1-\hat{p})}{R}} \quad (14)$$

To find the confidence interval, we use the standard error δ alongside a Z -score from the Standard Normal Distribution. Using the Wald formula, a 95% confidence interval (ci) can be calculated by:

$$ci = \hat{p} \pm 1.96 \sqrt{\frac{\hat{p}(1-\hat{p})}{R}} \quad (15)$$

The margin of error Δ is then given by:

$$\Delta = 1.96 \sqrt{\frac{\hat{p}(1-\hat{p})}{R}} \quad (16)$$

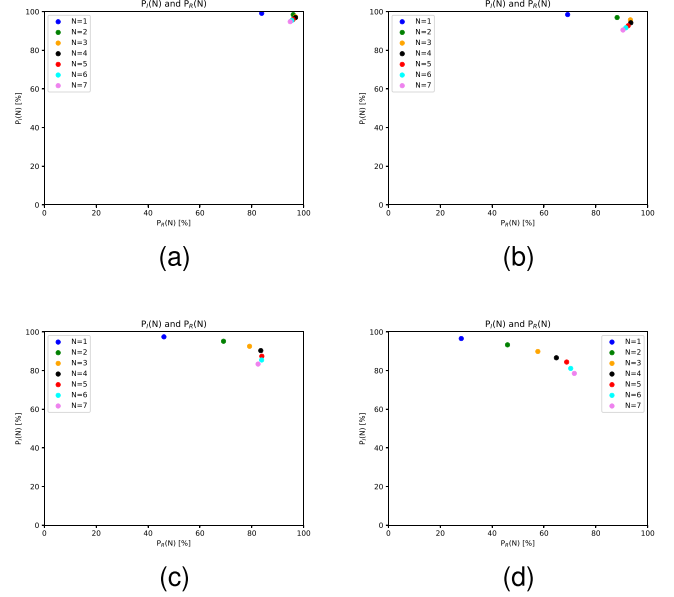


Fig. 5. Plots of $P_R(N)$ and $P_I(N)$ for $\beta = 0.005$ and $\gamma = 0.1$, and $1 \leq N \leq 7$, calculated by Monte Carlo simulation, with (a) $I_0 = 0.1$, (b) $I_0 = 0.2$, (c) $I_0 = 0.4$, and (d) $I_0 = 0.6$.

Table I compares $P_R(N)$ values from simulation with those calculated using (12). The findings indicate that the theoretical values and Monte Carlo results align within the margin of error Δ , confirming the reliability of the computational analysis.

In the next two subsections, we demonstrate how the parameters β , γ , and I_0 affect the probabilities $P_R(N)$ and $P_I(N)$. Values for β , γ , and I_0 were obtained from preliminary simulations to generate a diverse set of $P_R(N)$ and $P_I(N)$ distributions, ranging from tightly clustered data points to smooth, curve-like distributions with varying degrees of curvature.

B. For $\gamma = 0.1$ and an Initial Fraction of Infected $I_0 = 0.1, 0.2, 0.4$, and 0.6

Here, we keep $\gamma = 0.1$ constant and evaluate the impact of β on the system dynamics by simulating with $\beta = 0.005, 0.01$, and 0.02 .

1) For $\beta=0.005$: The probabilities $P_I(N)$ and $P_R(N)$ for $I_0 = 0.1, 0.2, 0.4$, and 0.6 are shown in Fig. 5(a), (b), (c), and (d), respectively. According to the plots, for small number of nodes N , $P_R(N)$ is significantly reduced with increasing I_0 , whereas $P_I(N)$ is much less affected by the initial fraction of infected. In addition, by raising I_0 , the curve is stretched

TABLE II
OPTIMUM NUMBER OF NODES N THAT MAXIMIZES $P_R(N)$ FOR $\beta = 0.005$
AND $\gamma = 0.1$, WITH DIFFERENT VALUES OF I_0

I_0	0.1	0.2	0.4	0.6
$\text{MAX}(P_R(N))$ [%]	96.9	93.5	83.9	71.7
N	4	4	5	7

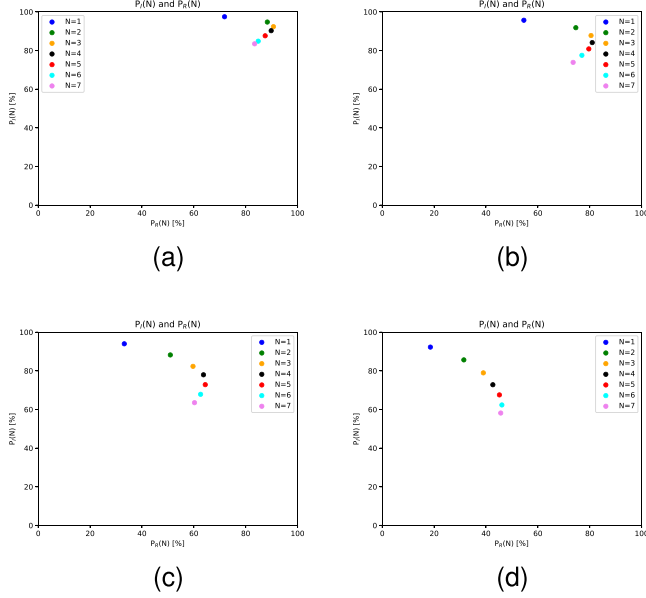


Fig. 6. Plots of $P_R(N)$ and $P_I(N)$ for $\beta = 0.01$ and $\gamma = 0.1$, and $1 \leq N \leq 7$, calculated by Monte Carlo simulation, with (a) $I_0 = 0.1$, (b) $I_0 = 0.2$, (c) $I_0 = 0.4$, and (d) $I_0 = 0.6$.

from strongly localized data points towards a smooth curved line.

Particularly, for $I_0 = 0.1$, and $I_0 = 0.2$, after reaching a maximum P_R at $N=4$, the data points closely approximate the boundary line given by $P_I(N) = P_R(N)$. Additionally, the simulation reveals that the number of nodes N that maximizes $P_R(N)$ increases with I_0 , as illustrated in Table II.

2) For $\beta = 0.01$: By doubling the value of β , namely, the probability of getting infected, the epidemic spreads significantly and consequently reduces the probability of the local network remaining free of infection, $P_I(N)$, as shown in Fig. 6.

The increase in β directly influences both P_R and P_I , shifting the data points downward and to the left. Furthermore, similarly to Fig. 5, the data points form an arc pointing to the right, as illustrated in Fig. 6.

For $I_0 = 0.1$, and $I_0 = 0.2$, after reaching a maximum P_R at $N = 3$ and $N = 4$, respectively, the data points closely approximate the boundary line given by $P_I(N) = P_R(N)$. In addition, increasing β from 0.005 to 0.01, notably reduces the maximum value of $P_R(N)$, especially for $I_0 \geq 0.2$, as shown in Table III.

3) For $\beta = 0.02$: By doubling the value of β from 0.01 to 0.02, the shape in Fig. 7 becomes more elongated, approaching the form of an arc pointed towards the right. Also, the reduction in P_R and P_I is a consequence of the enhanced spreading and increased risk of infecting the local network.

TABLE III
OPTIMUM NUMBER OF NODES N THAT MAXIMIZES $P_R(N)$ FOR $\beta = 0.01$
AND $\gamma = 0.1$, WITH DIFFERENT VALUES OF I_0

I_0	0.1	0.2	0.4	0.6
$\text{MAX}(P_R(N))$ [%]	90.7	81.0	64.4	46.1
N	3	4	5	6

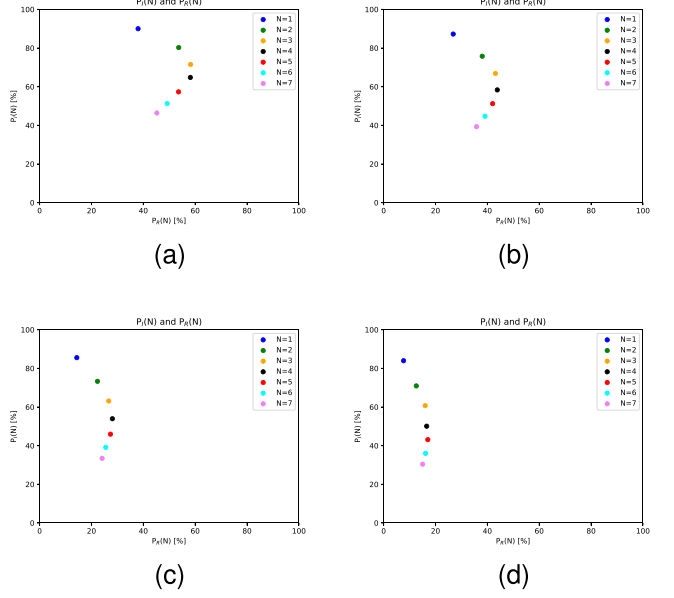


Fig. 7. Plots of $P_R(N)$ and $P_I(N)$ for $\beta = 0.02$ and $\gamma = 0.1$, and $1 \leq N \leq 7$, calculated by Monte Carlo simulation, with (a) $I_0 = 0.1$, (b) $I_0 = 0.2$, (c) $I_0 = 0.4$, and (d) $I_0 = 0.6$.

TABLE IV
OPTIMUM NUMBER OF NODES N THAT MAXIMIZES $P_R(N)$ FOR $\beta = 0.02$
AND $\gamma = 0.1$, WITH DIFFERENT VALUES OF I_0

I_0	0.1	0.2	0.4	0.6
$\text{MAX}(P_R(N))$ [%]	58.2	43.8	28.1	17.0
N	3	4	4	5

Similarly to the previous case, the increase in β also significantly reduces the maximum probability $P_R(N)$, as presented in Table IV.

C. For $\beta = 0.01$ and an Initial Fraction of Infected $I_0 = 0.1, 0.2, 0.4$, and 0.6

Here, we keep $\beta = 0.01$ constant and evaluate the impact of γ on the system dynamics by simulating with $\gamma = 0.05$, and 0.2 .

1) For $\gamma = 0.05$: The plots of $P_R(N)$ and $P_I(N)$ for $\beta = 0.01$ and $\gamma = 0.05$ are shown in Fig. 8. These plots are remarkably similar to those for $\beta = 0.02$ and $\gamma = 0.1$, shown in Fig. 7, demonstrating an excellent agreement between the data points for $I_0 = 0.1, 0.2, 0.4$, and 0.6 .

This results from the fact that the two cases exhibit a comparable dynamic behavior; that is, the curves of the number of susceptible (S), infected (I), and removed (R) individuals over 300 timesteps are numerically similar, as presented in Fig. 9. Indeed, by doubling both β from 0.01 to 0.02, and γ from 0.05 to 0.1, increase by the same proportion the probabilities of being

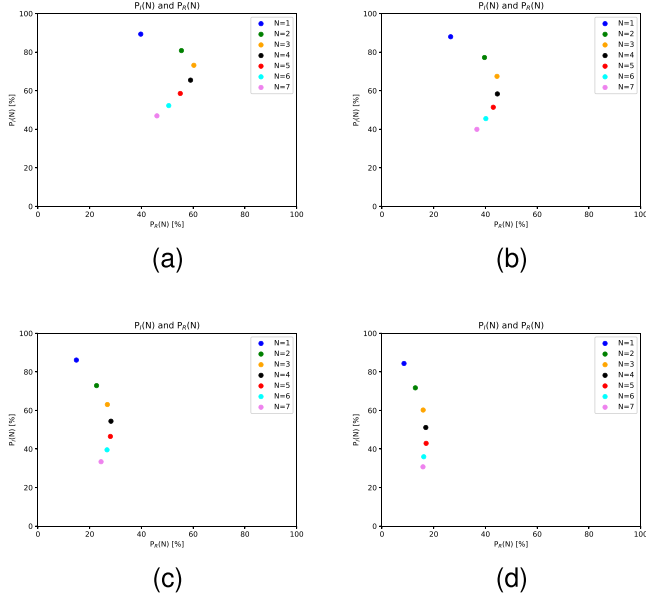


Fig. 8. Plots of $P_R(N)$ and $P_I(N)$ for $\gamma = 0.05$ and $\beta = 0.01$, and $1 \leq N \leq 7$, calculated by Monte Carlo simulation, with (a) $I_0 = 0.1$, (b) $I_0 = 0.2$, (c) $I_0 = 0.4$, and (d) $I_0 = 0.6$.

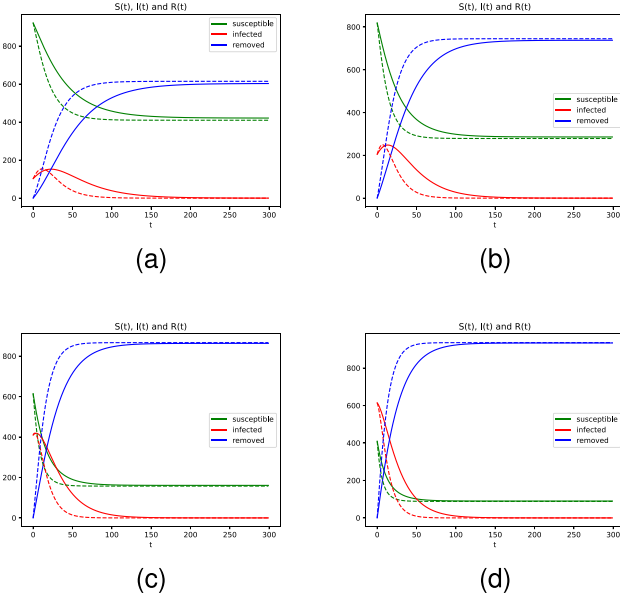


Fig. 9. Comparison of $S(t)$, $I(t)$ and $R(t)$ for $\beta = 0.01$ and $\gamma = 0.05$ (solid lines), and $\beta = 0.02$ and $\gamma = 0.1$ (dashed lines), with (a) $I_0 = 0.1$, (b) $I_0 = 0.2$, (c) $I_0 = 0.4$, and (d) $I_0 = 0.6$.

infected and getting removed, which have opposing effects on the spreading: the former enhances the propagation, whereas the latter mitigates the epidemic by shortening the time the cell is allowed to infect.

From the aforementioned discussion, we conclude that reducing γ by 50%, from 0.1 to 0.05, and keeping $\beta = 0.01$ produces a similar effect to those of doubling β from 0.01 to 0.02 and keeping $\gamma = 0.1$. Furthermore, these facts are also consistent with the resemblance between the optimum number of edges N

TABLE V
OPTIMUM NUMBER OF NODES N THAT MAXIMIZES $P_R(N)$ FOR $\beta = 0.01$ AND $\gamma = 0.05$, WITH DIFFERENT VALUES OF I_0

I_0	0.1	0.2	0.4	0.6
$\text{MAX}(P_R(N))$ [%]	60.3	44.6	28.2	17.1
N	3	4	4	5

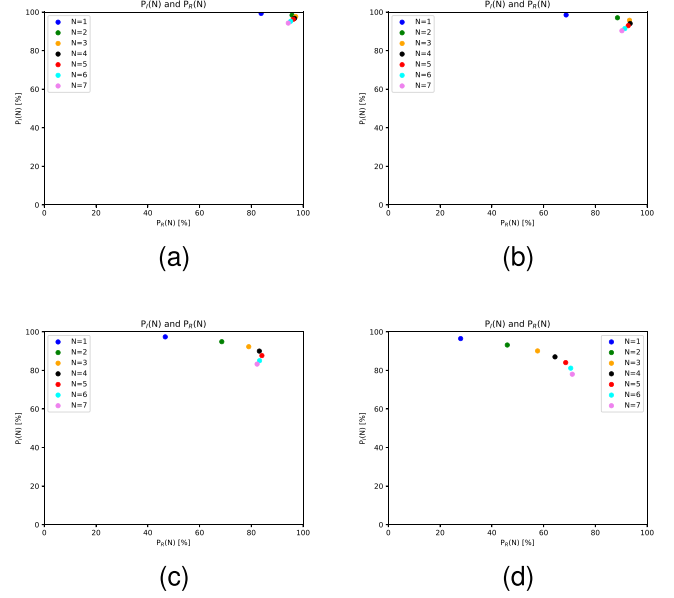


Fig. 10. Plots of $P_R(N)$ and $P_I(N)$ for $\gamma = 0.2$ and $\beta = 0.01$, and $1 \leq N \leq 7$, calculated by Monte Carlo simulation, with (a) $I_0 = 0.1$, (b) $I_0 = 0.2$, (c) $I_0 = 0.4$, and (d) $I_0 = 0.6$.

TABLE VI
OPTIMUM NUMBER OF NODES N THAT MAXIMIZES $P_R(N)$ FOR $\beta = 0.01$ AND $\gamma = 0.2$, WITH DIFFERENT VALUES OF I_0

I_0	0.1	0.2	0.4	0.6
$\text{MAX}(P_R(N))$ [%]	97.2	93.5	84.1	71.1
N	3	4	5	7

for $\beta = 0.01$ and $\gamma = 0.05$, shown in Table V, and for $\beta = 0.02$, and $\gamma = 0.1$, presented in Table IV.

2) For $\gamma = 0.2$: The plots of $P_R(N)$ and $P_I(N)$ for $\beta = 0.01$ and $\gamma = 0.2$ are shown in Fig. 10. These plots have a great similarity with those for $\beta = 0.005$ and $\gamma = 0.1$, shown in Fig. 5, demonstrating an outstanding agreement between the data points for $I_0 = 0.1, 0.2, 0.4$, and 0.6 . The excellent correspondence between the two cases can be understood as a consequence of the similar dynamic behavior, represented by the curves of susceptible $S(t)$, infected $I(t)$, and removed $R(t)$ for the two conditions, as shown in Fig. 11.

Therefore, analogously to the previous case, in which reducing γ by 50% produced comparable results to increasing β by 100%, here, increasing γ by a factor of two yielded similar outcomes to reducing β by 50%. Besides, these results also explain the close agreement between the optimum number of edges N for $\beta = 0.01$ and $\gamma = 0.2$, shown in Table VI, and for $\beta = 0.005$ and $\gamma = 0.1$, presented in Table II.

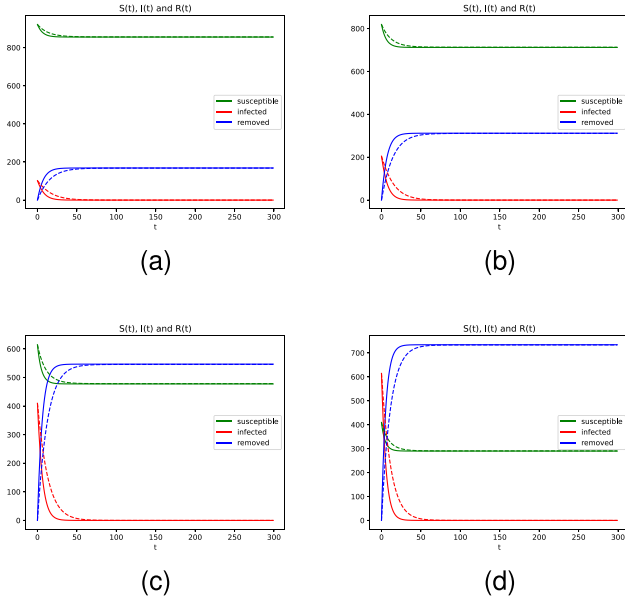


Fig. 11. Comparison of $S(t)$, $I(t)$ and $R(t)$ for $\beta = 0.01$ and $\gamma = 0.2$ (solid lines), and $\beta = 0.005$ and $\gamma = 0.1$ (dashed lines), with (a) $I_0 = 0.1$, (b) $I_0 = 0.2$, (c) $I_0 = 0.4$, and (d) $I_0 = 0.6$.

TABLE VII
OPTIMUM NUMBER OF NODES N THAT MAXIMIZES $P_R(N)$ FOR DIFFERENT
VALUES OF β , γ AND I_0

β	I_0	$\gamma = 0.05$	$\gamma = 0.1$	$\gamma = 0.2$
0.005	0.1	N=3	N=4	N=3
	0.2	N=4	N=4	N=4
	0.4	N=4	N=5	N=5
	0.6	N=6	N=7	N=7
0.01	0.1	N=3	N=3	N=3
	0.2	N=4	N=4	N=4
	0.4	N=4	N=5	N=5
	0.6	N=5	N=6	N=7
0.02	0.1	N=3	N=3	N=3
	0.2	N=3	N=4	N=4
	0.4	N=4	N=4	N=5
	0.6	N=3	N=5	N=6

D. Practical Applicability and Final Remarks

Table VII summarizes simulation results for different values of β , γ , and I_0 , highlighting their impact on the optimal number of edges, N .

It is important to note that significant changes in parameters like β , γ , or I_0 have little effect on the optimal number of edges, N . This shows that the method for selecting the optimal number of links remains effective even in real-world settings where such values are imprecise and often challenging to estimate.

It is also worth mentioning that, although the multilayer network has many nodes, with 1024 in each layer, the optimal number of edges is much fewer. This demonstrates a significant, outstanding result of this work: the security and robustness of large multilayer networks can be optimized with only a few interlayer edges.

Multilayer networks have experienced significant growth and widespread adoption across various areas, including smart

homes, intelligent transportation, and critical infrastructure systems such as power transmission lines. The Internet of Things (IoT) is a prominent example of multilayer infrastructure in these domains, with the device network at the bottom and the cellular communication network supporting cloud computing [12].

Therefore, the method described in this work for identifying the optimal number of interlayer edges is appropriate for practical use and can guide the design of secure and robust IoT networks or enterprise communication systems, especially those in which malware behavior follows the SIR (susceptible-infected-removed) model.

E. Optimal Connectivity of Smart Grid Infrastructure: An Application Example

Here, to demonstrate how to use the model to help design a robust and secure complex network within a realistic context, we examine a hypothetical scenario of a cyberattack against a smart grid. In this infrastructure, a wireless sensor network (WSN) collects field data and transmits it via cellular network to a supervisory-control and data-acquisition (SCADA) system for decision and monitoring purposes.

The WSN is composed of sensor and sink nodes, where a sensor node acquires data and sends it through multi-hop relays, eventually reaching a sink node, which uploads the data from the WSN to the SCADA network. The SCADA and WSN are depicted as a two-layer network connected by interlayer edges, representing the communication links between the sink nodes in the WSN and the SCADA network.

In the analysis, both the WSN and the SCADA are represented as cellular automata. Besides, cellular automata are suitable for describing electric power grids due to their particular topological characteristics: they are strongly localized with many connections across physically neighboring nodes [20]. In fact, cellular automata have already been employed to simulate cyberattacks against smart grids [15], [21]. Similarly, cellular automata have shown to be capable of properly representing malware spreading on WSNs [22], [23].

In the model, we consider the following assumptions:

- 1) Both the SCADA and the WSN layers have a grid of 32 x 32 cells, with a Moore neighborhood;
- 2) While the SCADA layer consists of stationary, uniform cells, the WSN layer contains cells occupied by either a sensor or a sink node;
- 3) Every sink node in the WSN layer is connected through an interlayer edge to a node in the SCADA layer;
- 4) There are N interlayer edges, which represent the cellular network links that transmit data from the sink nodes in WSN to the SCADA;
- 5) The SIR epidemic model describes the malware propagation.

Fig. 12 illustrates a two-layer architecture comprising a SCADA network and a WSN. The edges between these layers represent cellular communication linking the WSN sink nodes to the SCADA.

In this scenario, a malware initially deployed to the WSN propagates by jumping from sensor to sensor until it reaches a

TABLE VIII
 $P_R(N)$ AND CORRESPONDING MARGIN OF ERROR (Δ) FOR DIFFERENT VALUES OF N AND β

N	1	2	3	4	5	6	7
$P_R(N)$ ($\beta = 0.45\%$)	35.93%	46.65%	48.41%	44.49%	39.89%	35.12%	31.01%
Δ	0.77%	0.80%	0.80%	0.80%	0.78%	0.76%	0.74%
$P_R(N)$ ($\beta = 0.50\%$)	26.26%	34.61%	36.39%	33.22%	29.78%	24.65%	20.27%
Δ	0.70%	0.76%	0.77%	0.75%	0.73%	0.69%	0.64%

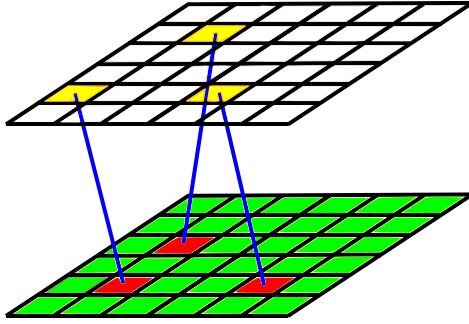


Fig. 12. The two-layer architecture integrates a SCADA (upper level) with a WSN (lower level). Green squares represent sensor nodes, while red squares denote sink nodes within the WSN. Blue edges illustrate the communication links bridging the sink nodes (red) to the SCADA nodes (yellow).

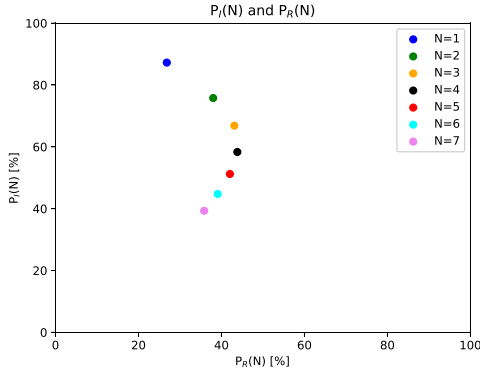


Fig. 13. Plots of $P_R(N)$ and $P_I(N)$, for $\beta = 0.02$, $\gamma = 0.1$, and $I_0 = 0.2$, with $1 \leq N \leq 7$.

sink node. Once infected, the sink node may either transmit the malware to the SCADA network or fail, disabling its gateway communication capability. In particular, if all sink nodes fail, the WSN becomes completely isolated from the digital infrastructure, resulting in severe consequences for the smart grid.

Accordingly, designing a WSN is a balancing problem. While adding sink nodes reduces the risk of network isolation, it facilitates malware propagation from the WSN to the SCADA. Thus, planning a robust and secure network involves finding the optimal trade-off between these two terms by determining the number of sink nodes N that maximizes $P_R(N)$.

In this example, the network is expected to withstand a cyber-attack that initially compromises 2% of the nodes in the WSN. We assume that infected nodes are removed with probability $\gamma = 1\%$. In addition, we estimate that susceptible nodes are infected with a probability of β , ranging from 0.45% to 0.50%.

The program ran 15,000 Monte Carlo simulations to determine $P_R(N)$ across two conditions where β varied (0.45% and 0.50%), holding I_0 at 2% and γ at 1%. Table VIII summarizes the simulation results, with respective margins of error (Δ) for different values of β .

The analysis demonstrated that across the two β scenarios (0.45% and 0.50%), the maximum value of $P_R(N)$ occurs at $N = 3$. Therefore, in this case, deploying a WSN with three sink nodes is the optimal configuration for ensuring a robust and secure network infrastructure.

It should be emphasized that, for simplicity, we assumed that sensor and sink nodes exhibit identical functional behavior and can be represented by the same SIR model. Furthermore, the optimality of the three-sink nodes configuration does not imply that all spatial distributions perform equally. Positioning sink nodes in close proximity results in different robustness levels than spreading them across the network area. However, three sink nodes represent the optimal quantity when distributed randomly across the network.

V. A METHOD FOR CHOOSING THE OPTIMUM NUMBER OF EDGES

In this section, we present a methodology for selecting the optimal number of edges, N , while considering two parameters that depend on preference: $P_I(N)$ and $P_R(N)$.

Although finding the value of N that maximizes P_R is a straightforward manner for choosing the optimum number of internetwork edges, this approach does not take into consideration another important parameter, P_I , which governs the probability of the local network remaining free of infection.

Additionally, selecting N based solely on P_R may lead to an inappropriate decision. A particular case that illustrates this concept is presented in Section IV-B3, for $\beta = 0.02$, $\gamma = 0.1$ and $I_0 = 0.2$, and shown in Fig. 13, where the maximum P_R , which occurs for $N = 4$, and the second largest, for $N = 3$, are very close, and are equal to $43.98 \pm 0.25\%$ and $43.09 \pm 0.25\%$, respectively. Nevertheless, the value of P_I for $N = 4$ ($58.60 \pm 0.25\%$) is much less than for $N = 3$ ($66.6 \pm 0.24\%$). Thus, choosing $N = 4$ over $N = 3$ results in only a slightly greater P_R but an appreciable reduction in P_I . Here, we simulated the same scenario but increased the number of Monte Carlo runs to 150,000 to improve precision and reduce the associated margin of error.

To accurately determine the optimum number of internetwork edges, we introduce an approach based on the weighted scoring method, which considers P_I and P_R and incorporates preference by assigning weights to each parameter. This approach involves selecting the number of edges N that maximizes the following

TABLE IX
VALUES OF $P_R(N)$, $P_I(N)$, AND $f(N)$, ALONG WITH CORRESPONDING MARGINS OF ERROR

N	1	2	3	4	5	6	7
$P_R(N)$	25.67%	38.33%	43.09%	43.98%	42.01%	39.00%	35.57%
Δ	0.22%	0.25%	0.25%	0.25%	0.25%	0.25%	0.24%
$P_I(N)$	87.52%	76.41%	66.6%	58.6%	51.04%	44.61%	39.05%
Δ	0.17%	0.21%	0.24%	0.25%	0.25%	0.25%	0.25%
$f(N)$	56.60%	57.37%	54.85%	51.29	46.53%	41.81%	37.31%
Δ	0.20%	0.23%	0.25%	0.25%	0.25%	0.25%	0.25%

equation.

$$\max_N f(N) = x \cdot P_R(N) + (1 - x) \cdot P_I(N) \quad (17)$$

where, the parameter x , ($0 \leq x \leq 1$), express the preference towards P_R or P_I .

To illustrate the effectiveness of the proposed method, we determine N for the case presented in Section IV-B3, as shown in Fig. 13, using $x = 0.5$ for the preference parameter. The results for $f(N)$ are presented in Table IX.

From Table IX, we find that the optimum number of edges N , in other words, the value of N at which $f(N)$ is maximum, is $N=2$.

Additionally, we should note that the results may vary, as they are influenced by personal preferences, particularly the relative importance assigned to parameters P_I and P_R . If one places more value on P_I , the calculation will naturally favor options that score higher in that area, even if those options perform worse on P_R . Accordingly, the optimal number N is subjective and can change depending on the preference, represented by the parameter x .

VI. CONCLUSION

Complex and dedicated networks involving multilayer topologies are becoming increasingly popular and have gained significant importance in modern technological applications. Thus, studying the propagation of malware over multilayer networks is critical for both predicting infection and designing resilient architectures.

In this paper, we introduce probabilistic measures of network security and robustness and demonstrate that there exists a number of interlayer edges that balances them. We have studied the impact of the number of interconnections N , on the vulnerability and robustness of multilayer networks. Specifically, we introduced a two-layer framework to investigate the problem at two levels: an external level and a local network level. We examined the problem analytically for the probability of infecting $\beta = 0$ and numerically, using the Monte Carlo method, under different parameters β , γ and I_0 of the propagation model. According to our simulations and analysis, the results are summarized below:

- 1) For the particular case of $\beta = 0$, the probability of infecting the local network $P_I(N) = 0$, whereas the probability of at least one edge remaining intact $P_R(N)$ increases with N ;

- 2) For $\beta > 0$, there is an optimal number of edges N that maximizes the probability of remaining connected while not infecting the local network;
- 3) Increasing β , enhances the epidemic spreading, reducing both $P_I(N)$ and $P_R(N)$;
- 4) Increasing the probability of being removed γ has similar effects to reducing β . This similarity is due to the fact that the increase in β and γ have opposing effects on the spreading: the former enhances the propagation, whereas the latter mitigates the epidemic by shortening the time the cell can infect;
- 5) The initial fraction of infected I_0 has a significant impact on both P_I and P_R , as enhancing the number of infected cells appreciably increases the probability of infecting the local network, thereby reducing P_I and P_R .

This paper also offers an alternative understanding of the dynamical behavior of multilayer networks, framing it as a phase transition occurring at a critical threshold. Moreover, a significant accomplishment of this work is the demonstration that the SIR compartmental epidemic model effectively encapsulates the competing notions of security and robustness within multilayer networks.

In addition, this work introduces an approach for assessing the optimal number of internetwork edges that takes into account the probabilities P_I , P_R , and personal preferences. This method yields more accurate results than considering only P_R , as the precise decision should also involve subjective appraisals.

Further research on this subject may include the study of the impact of connectivity on more realistic topologies, such as those with many connections across multiple layers. Examining topologies that better represent real-life problems with complex geometries provides significant advantages for understanding, predicting, and mitigating cyber threats. Besides, since mathematical models, such as the SIR model adopted in this work, typically rely on simplified assumptions, simulating on realistic topologies helps validate or fine-tune the theoretical representation.

Finally, while this study provides a numerical characterization of robustness and vulnerability, future investigations could focus on deriving analytical thresholds and bifurcation conditions to formally define the phase transitions of inter-layer connectivity.

REFERENCES

- [1] G. Avoine and J. Hernandez-Castro, *Security of Ubiquitous Computing Systems: Selected Topics.*, Berlin, Germany: Springer, Jan. 2021.

- [2] S. Karageorgiou and V. Karyotis, "Markov-based malware propagation modeling and analysis in multi-layer networks," *Network*, vol. 2, pp. 456–478, Sep. 2022.
- [3] M. Salehi, R. Sharma, M. Marzolla, M. Magnani, P. Siyari, and D. Montesi, "Spreading processes in multilayer networks," *IEEE Trans. Netw. Sci. Eng.*, vol. 2, no. 2, pp. 65–83, Apr.–Jun. 2015.
- [4] X. Wang, A. Aleta, D. Lu, and Y. Moreno, "Directionality reduces the impact of epidemics in multilayer networks," *New J. Phys.*, vol. 21, Sep. 2019, Art. no. 093026.
- [5] V. Karyotis, "A Markov random field framework for modeling malware propagation in complex communications networks," *IEEE Trans. Dependable Secure Comput.*, vol. 16, no. 4, pp. 551–564, Jul./Aug. 2019.
- [6] J. Chen, S. Sun, C. Xia, D. Shi, and G. Chen, "Modeling and analyzing malware propagation over wireless networks based on hypergraphs," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 6, pp. 3767–3778, Nov./Dec. 2023.
- [7] S. Zhou, S. Xu, L. Wang, Z. Liu, G. Chen, and X. Wang, "Propagation of interacting diseases on multilayer networks," *Phys. Rev. E*, vol. 98, Jul. 2018, Art. no. 012303.
- [8] M. Dickison, S. Havlin, and H. E. Stanley, "Epidemics on interconnected networks," *Phys. Rev. E - Stat., Nonlinear, Soft Matter Phys.*, vol. 85, Jun. 2012, Art. no. 066109.
- [9] A. Saumell-Mendiola, M. A. Serrano, and M. Boguñá, "Epidemic spreading on interconnected networks," *Phys. Rev. E - Stat., Nonlinear, Soft Matter Phys.*, vol. 86, Aug. 2012, Art. no. 026106.
- [10] C.-L. Chen, Q. P. Zheng, A. Veremyev, E. L. Pasilião, and V. Boginski, "Failure mitigation and restoration in interdependent networks via mixed-integer optimization," *IEEE Trans. Netw. Sci. Eng.*, vol. 8, no. 2, pp. 1293–1304, Apr.–Jun. 2021.
- [11] P. K. Huynh, M. M. Rahman, O. P. Yadav, T. Q. Le, and C. Le, "Assessing robustness and vulnerability in interdependent network infrastructure: A multilayer network approach," in *Proc. Annu. Rel. Maintainability Symp.*, 2024, pp. 1–7.
- [12] J. Chen, C. Touati, and Q. Zhu, "Optimal secure two-layer IoT network design," *IEEE Trans. Control Netw. Syst.*, vol. 7, no. 1, pp. 398–409, Mar. 2020.
- [13] O. Yagan, D. Qian, J. Zhang, and D. Cochran, "Optimal allocation of interconnecting links in cyber-physical systems: Interdependence, cascading failures, and robustness," *IEEE Trans. Parallel Distrib. Syst.*, vol. 23, no. 9, pp. 1708–1720, Sep. 2012.
- [14] D. Duan, X. Yang, C. Lv, and Z. Cai, "Modeling and predicting malware propagation in double-layer computer networks," *IEEE Trans. Netw. Sci. Eng.*, vol. 13, pp. 717–727, 2026.
- [15] A. C. B. Godoi and J. R. C. Piqueira, "Spatio-temporal malware containment model with alert," *Chaos, Solitons Fractals*, vol. 173, 2023, Art. no. 113618. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0960077923005192>
- [16] Y. Song, G.-P. Jiang, and Y. Gu, "Modeling malware propagation in complex networks based on cellular automata," in *Proc. IEEE Asia Pacific Conf. Circuits Syst.*, 2008, pp. 259–263.
- [17] D. Grattarola, L. Livi, and C. Alippi, "Learning graph cellular automata," in *Proc. 35th Int. Conf. Neural Inf. Process. Syst.*, 2021, pp. 20983–20994.
- [18] N. Bacaër, "McKendrick and Kermack on epidemic modelling (1926–1927)," in *A Short Hist. of Math. Popul. Dyn.*, London, U.K.: Springer, 2011, pp. 89–96, doi: [10.1007/978-0-85729-115-8_16](https://doi.org/10.1007/978-0-85729-115-8_16).
- [19] C. F. Bolz, A. Cuni, M. Fijalkowski, and A. Rigo, "Tracing the meta-level: PyPy's tracing JIT compiler," in *Proc. 4th Workshop Implementation, Compilation, Optim. Object-Oriented Lang. Program. Syst.*, 2009, pp. 18–25, doi: [10.1145/1565824.1565827](https://doi.org/10.1145/1565824.1565827).
- [20] S. F. Bush, "Network theory and smart grid distribution automation," *IEEE J. Sel. Areas Commun.*, vol. 32, no. 7, pp. 1451–1459, Jul. 2014.
- [21] G. Cisotto and L. Badia, "Cyber security of smart grids modeled through epidemic models in cellular automata," in *Proc. IEEE 17th Int. Symp. World Wireless, Mobile Multimedia Netw.*, 2016, pp. 1–6.
- [22] Z. He, Y. Lin, Y. Liang, X. Wang, A. M. Vera VenkataSai, and Z. Cai, "Modeling malware propagation dynamics and developing prevention methods in wireless sensor networks," in *Nonlinear Combinatorial Optimization.*, Cham, Switzerland: Springer, 2019, pp. 231–250, doi: [10.1007/978-3-030-16194-1_10](https://doi.org/10.1007/978-3-030-16194-1_10).
- [23] H. Zhou, S. Shen, and J. Liu, "Malware propagation model in wireless sensor networks under attack–defense confrontation," *Comput. Commun.*, vol. 162, pp. 51–58, 2020. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0140366420318880>

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